



Matrix Theory

17 Gram-Schmidt Orthogonalization

Dr. Konda Reddy Mopuri
Dept. of AI, IIT Hyderabad
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Orthonormal vectors



భారతీయ సాంకేతిక విజ్ఞాన సంస్థ హైదరాబాద్
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The vectors $q_1, q_2, \dots, q_n$ are orthonormal if

$$q_i^T q_j = \begin{cases} 1, & \text{whenever } i = j \\ 0, & \text{whenever } i \neq j \end{cases}$$

Orthonormal vectors



- A matrix with orthonormal columns is represented with Q

$$Q = \begin{bmatrix} | & | & & | \\ v_1 & v_2 & \dots & v_n \\ | & | & & | \end{bmatrix}$$

Orthonormal vectors

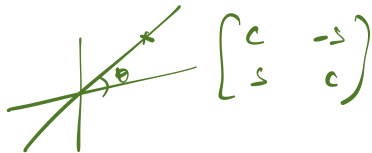


- A matrix with orthonormal columns is represented with Q
- Example: standard basis

$$Q_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$Q_n = \begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & & 0 \\ 0 & 0 & & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & & 1 \end{bmatrix}$$

Orthonormal basis



$$P \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} z \\ y \\ x \end{pmatrix}$$

what about a rotation matrix?

what about the permutation matrix?

- Is this the only orthonormal basis?

$$P = \frac{aa^T}{a^T a} x$$

$$\rightarrow \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ 0 \end{pmatrix}$$

what about the projection matrix?

And a reflection matrix?

Orthogonal Matrix



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An orthogonal matrix is a square matrix with orthonormal columns.

Orthogonal Matrix



An orthogonal matrix is a square matrix with orthonormal columns.

- If Q has orthonormal columns, then $Q^T Q = I$

$$\begin{bmatrix} -q_1^T \\ -q_2^T \\ \vdots \\ -q_n^T \end{bmatrix} \begin{bmatrix} | & & | \\ q_1 & q_2 & q_n \\ | & & | \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ \vdots & & \ddots & \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Orthogonal Matrix



An orthogonal matrix is a square matrix with orthonormal columns.

- If Q has orthonormal columns, then $Q^T Q = I$
- For a square orthogonal matrix, $Q^T = Q^{-1}$

we know for square
matrices the left & right
inverses are the same.

$$Q^T Q = I \Rightarrow Q Q^T = I$$
$$\Rightarrow Q^T = Q^{-1}$$

Orthogonal Matrix



Rotation

- Example: $Q = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$

Orthogonal Matrix



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permutation

- Example: $P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$

Orthogonal Matrix

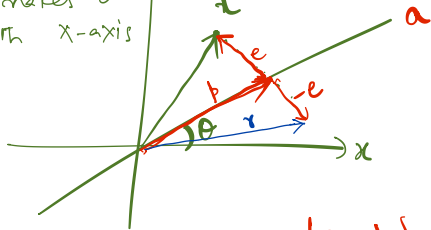


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x - given point (vector)

$\bar{a}$ - reference line
that makes θ
with x -axis

r - reflection of x
w.r.t. $\bar{a}$



• Example: $Ref = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

$$p = \frac{a a^T}{a^T a} x$$

$$x = e + p$$

$$e = (x - p)$$

$$(p - x) + p$$

$$= 2p - x$$

p is the projection of x on a

$$Ref = \left[\frac{2aa^T}{a^T a} - I \right] x$$

Orthogonal Matrix preserves length



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- $\bullet \|Qx\| = \|x\|$

rotation,
reflection,
permutation
preserve length

$$\sqrt{x^2 + y^2 + z^2}$$

$$\begin{matrix} 1 & 2 \\ y & y \\ z & x \end{matrix}$$

$$\sqrt{z^2 + y^2 + x^2}$$

$$\begin{aligned} \|Qx\| &= (Qx)^T Qx \\ &= x^T Q^T Qx \\ &= x^T x \\ &= \|x\| \end{aligned}$$

Orthonormal basis



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- Any vector is a combination of the basis vectors

Orthonormal basis



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- Any vector is a combination of the basis vectors
- This is exceptionally simple for an orthonormal basis

Orthonormal basis



$\mathbb{R}^n$

$$\mathcal{Q} = \{ q_1, q_2, \dots, q_n \}$$

Write b as a combination $b = x_1 q_1 + x_2 q_2 + \dots + x_n q_n$

Orthonormal basis



Write b as a combination $b = x_1q_1 + x_2q_2 + \dots + x_nq_n$

- How to find the coefficients?

$$v_i^T b = x_1 v_i^T v_1 + 0 + 0 + \dots$$
$$v_i^T b = x_i$$

$\Downarrow$
1

Orthonormal basis



Write b as a combination $b = x_1q_1 + x_2q_2 + \dots + x_nq_n$

- How to find the coefficients?
- Each piece of b has a simple formula, and recombining the pieces gives back b : $b = (q_1^T b)q_1 + (q_2^T b)q_2 + \dots + (q_n^T b)q_n$

Every vector b is the summation of its 1D projections onto the orthonormal basis $[q_i]$.
Since these projections are orthogonal,
$$\|b\|^2 = (q_1^T b)^2 + (q_2^T b)^2 + \dots + (q_n^T b)^2$$

(pythagoras)

Rows of an orthogonal matrix



The rows of a square matrix are orthonormal whenever the columns are.

$$Q^T \text{ becomes } Q^{-1} \quad [\because Q^T Q = Q Q^T = I]$$

for $Q Q^T = I$, the rows of Q also needs to be orthonormal. Note that the rows point in different directions from columns, but they are orthonormal.

- We started with $Ax = b$ when A is rectangular & over-determined

when we solve the square system $Qx = b$

$Qx = b$ columns of Q are orthonormal basis

$$\Rightarrow \hat{x} = Q^T b = Q^T b$$

$\Rightarrow$ Components of the solution $\hat{x}$ are the inner products $v_i^T b$

- We started with $Ax = b$ when A is rectangular & over-determined
- What about $Qx = b$ ($Q_{m \times n}, m > n$)? (i.e., columns of Q are orthogonal)

when we solve the square system $Qx = b$

$Qx = b$ columns of Q are orthonormal basis

$\Rightarrow x = Q^T b = Q^T b$ exact and unique solution

$\Rightarrow$ Components of the solution $\hat{x}$ are the inner products $a_i^T b$

- We started with $Ax = b$ when A is rectangular
- What about $Qx = b$ ($Q_{m \times n}, m > n$)? no exact solution $\Rightarrow$ LS solution

However, the orthonormal columns should simplify the problem.

Normal equations of least-squares $\Rightarrow Q^T Q x = Q^T b$

$\Rightarrow \hat{x} = Q^T b$ (since $Q^T Q = I$; even in the rectangular case)

$\therefore \hat{x} = Q^T b$ is $\begin{cases} \text{exact solution if } Q \text{ is square \& orthogonal} \\ \text{LS solution if } Q \text{ is rectangular with orthonormal columns} \end{cases}$

Projecting onto orthonormal basis



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- Consider projecting (x, y, z) onto the xy plane

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} \rightarrow \begin{bmatrix} x \\ y \\ 0 \end{bmatrix} = \begin{bmatrix} x \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ y \\ 0 \end{bmatrix}$$

projection onto x -axis

projection onto y -axis

Projecting onto orthonormal basis



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$$y = C + Dt$$

$$t = -1, 0, 1$$

$$y_1 = C + (-1)D$$

$$y_2 = C + 0 \cdot D$$

$$y_3 = C + 1 \cdot D$$

$$\begin{pmatrix} 1 & -1 \\ 1 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} C \\ D \end{pmatrix} = \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix}$$

Q

- When the measurement times average to zero, fitting a straight line leads to orthogonal columns

$$\hat{C} = \frac{(1, 1, 1)^T (y_1, y_2, y_3)}{(1, 1, 1)^T (1, 1, 1)}$$

$$\hat{D} = \frac{(-1, 0, 1)^T (y_1, y_2, y_3)}{\sqrt{(-1)^2 + 0^2 + 1^2}}$$

we can project y separately onto each column

Projecting onto orthonormal basis



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- What if the measurement times don't average to zero?

time origin can be shifted to the mean $\bar{t} = \frac{1}{n} \sum_{i=1}^n t_i$

then instead of fitting $y = c + Dt$ we will

work with $y = c + D(t - \bar{t})$.

$$\hat{c} = \frac{(1, 1, \dots)^T (y_1, y_2, \dots, y_n)}{\sqrt{1^2 + \dots + 1^2}} \quad \hat{D} = \frac{(t_1 - \bar{t} \quad t_2 - \bar{t} \quad \dots \quad t_n - \bar{t}) (y_1, \dots, y_n)}{\sqrt{(t_1 - \bar{t})^2 + \dots + (t_n - \bar{t})^2}}$$

Projecting onto orthonormal basis



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Orthogonal columns are so much better that it is worth changing to that case.

The Gram-Schmidt Process



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- Suppose you are given three independent vectors a, b, c

The Gram-Schmidt Process



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- Suppose you are given three independent vectors a, b, c
- If they are orthonormal, life is easy

The Gram-Schmidt Process



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- Suppose you are given three independent vectors a, b, c
- If they are orthonormal, life is easy
- Procedure for making them orthonormal?

The Gram-Schmidt Process



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- Suppose you are given three independent vectors a, b, c
- If they are orthonormal, life is easy
- Procedure for making them orthonormal?
- Given a, b, c and we want q_1, q_2, q_3 .

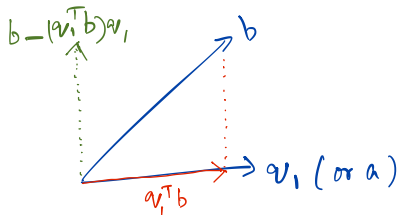
The Gram-Schmidt Process



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- $q_1 = \frac{a}{\|a\|}$ we can pick the first one and normalize it-

The Gram-Schmidt Process



- $q_1 = \frac{a}{\|a\|}$
- $B = b - (q_1^T b)q_1$ and $q_2 = \frac{B}{\|B\|}$

second one has to be orthogonal to the first.
so, we find its component in the direction of the first
and subtract it.

The Gram-Schmidt Process



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- $q_1 = \frac{a}{\|a\|}$
- $B = b - (q_1^T b)q_1$ and $q_2 = \frac{B}{\|B\|}$
- $C = c - (q_1^T c)q_1 - (q_2^T c)q_2$ and $q_3 = \frac{C}{\|C\|}$

repeat this procedure : remove the component
of C that is on the plane of v_1 & v_2

$A = QR$ Factorization



RHS expresses the original vector as a combination of the orthonormal basis

$$A = \begin{bmatrix} a & b & c \end{bmatrix} = \begin{bmatrix} q_1 & q_2 & q_3 \end{bmatrix} \begin{bmatrix} q_1^T a & q_1^T b & q_1^T c \\ q_2^T a & q_2^T b & q_2^T c \\ q_3^T a & q_3^T b & q_3^T c \end{bmatrix} = QR$$

A
initial set of
ind. vectors

Q
orthonormal
basis

R

↘ upper triangular
diagonal entries $\neq 0$
($\because a, b, c$ are independent)
 $\Rightarrow$ full rank matrix

(PTO)

Rough Work



QR decomposition simplifies the least squares
{ also known as } problem $Ax = b$
{ orthogonalization }

the normal equations become

$$A^T A x = A^T b$$

$$\Rightarrow (QR)^T QR x = (QR)^T b$$

$$R^T Q^T QR x = R^T Q^T b$$

$$R^T R x = R^T Q^T b$$

$$Rx = Q^T b \quad (\because R \text{ is full rank})$$

R is upper triangular

$\Rightarrow$ back substitution solves this